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Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

USE OF FENTANYL PCA FOR LABOUR ANALGESIA

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1. Introduction and Purpose

Fentanyl PCA is an useful option for labour analgesia when epidural analgesia is contraindicated; however it does not provide as effective analgesia. For safe use, both the parturient and neonate require careful monitoring both during labour and after delivery.

2. Scope

This guideline is intended as reference for trainee anaesthetists, and midwifes who wish to provide patient controlled fentanyl analgesia for parturients in labour.

3. Points to consider

3.1 Indication

Patient should be in the first stage of labour and not coping with the contraction pain despite analgesia with Entonox and intermittent intramuscular opioid injections. She must also be unable to have an epidural due to it being contraindicated (including maternal refusal).

3.2 Setting up the PCA

- The anaesthetist on call for delivery suite is responsible for prescribing the fentanyl PCA while the midwife is responsible for setting up and maintaining it.
- Use a dedicated Smiths Medical CADD Solis PCA device
- Draw up 500mcg fentanyl (10mls of 50mcg/ml) and dilute with 40 ml of saline in a 50 ml syringe to make up a 10 mcg/ml solution
- Fill the cassette with the 50ml of 10mcg/ml solution of fentanyl, and remove any air using the syringe, before clamping the cassette filling tube
- Label the PCA cassette clearly with the drug name, dilutant and concentration
- Load the cassette into the CADD Solis PCA device
- Program the CADD Solis PCA device as per standard fentanyl PCA prescription (see overleaf)
- Connect the extension tubing onto the cassette filling tube, unclamp the filling tube and prime the line

- Prescribe the PCA on the patient's prescription chart. The prescription should read :

  Fentanyl PCA (500mcg in 50 mls)
  Bolus dose : 10 mcg (1ml)
  Lockout period : 5 minutes

- Prescribe naloxone (100mcg intravenously every 2 minutes until respiratory rate > 8 /min and consciousness has returned or 400mcg intramuscularly if respiratory rate < 8 breaths /min)

- Connect the PCA to the patient using a Y- connector with a one way valve on the intravenous infusion port

3.3 Monitoring

- While the Fentanyl PCA is being used, the midwife should monitor the patients vital signs, particularly the respiratory rate and level of sedation, at regular intervals during labour

- If the midwife has any concern regarding the function of the Fentanyl PCA or it's effects on the patient, then she should contact the duty obstetric anaesthetist

3.4 Troubleshooting

- Beware of respiratory depression in the newborn baby due to the fentanyl, and have naloxone available. Always notify the paediatricians that the mother has received a Fentanyl PCA during labour

- If the patient's respiratory rate drops below 7 breath/min and/or they become unrousable then :

  i. SUMMON HELP, including calling the anaesthetist
  ii. Remember **A**irway **B**reathing **C**irculation
  iii. Give 6 litres oxygen via facemask if patient is still breathing. Alternatively, if the patient is apnoeic then ensure a patent airway then bag the patient with 100% oxygen using an Ambu Bag)
  iv. Disconnect the PCA device
  v. Give naloxone intravenously or intramuscularly

4. References

1. Fortescue C, Wee MYK. Analgesia in labour : non-regional techniques. *Br J Anaesthesia: CEACCP* (2005) Vol. 5 (1): 9-13